

Task 3: Car Price Prediction with Machine Learning

1. Data Preprocessing and Feature Engineering

To make the data suitable for a regression model, we performed several key steps:

- **Car Age Calculation:** Instead of using the raw "Year," we calculated the **Age of the Car** (Current Year - Year of Manufacture), as the age of a vehicle is a more direct indicator of its depreciation.
- **Categorical Encoding:** Features like **Fuel Type** (Petrol, Diesel, CNG), **Selling Type** (Dealer vs. Individual), and **Transmission** (Manual vs. Automatic) were converted into numerical values using One-Hot Encoding.
- **Feature Selection:** We focused on key drivers such as the **Present_Price** (original showroom price), **Driven_kms**, **Owner** history, and **Car_Age**.

2. Model Performance

We trained a **Random Forest Regressor**, an ensemble model that excels at capturing non-linear relationships between features.

Evaluation Results:

- **R-squared Score (R^2): 0.9595** (This means the model explains approximately 96% of the variance in car prices).
- **Mean Absolute Error (MAE): 0.637** (On average, the predictions are within 0.63 units of the actual price).

3. Key Price Drivers (Feature Importance)

The model identified the most influential factors in determining a car's resale value:

1. **Present Price:** The original cost of the car remains the strongest predictor of its resale value.
2. **Car Age:** As expected, older cars depreciate significantly.
3. **Driven Kilometers:** Higher mileage generally correlates with lower prices.
4. **Transmission Type:** Automatic cars often command a higher resale premium in the used market.

4. Real-World Applications

Machine learning in price prediction is widely used across various industries:

- **E-commerce & Used Car Platforms:** Companies like CarDekho, Cars24, or eBay use similar models to provide instant valuations to sellers and fair pricing to buyers.
- **Real Estate:** Predicting property values based on location, square footage, and neighborhood amenities.
- **Stock Market:** Estimating future asset prices based on historical trends and economic indicators.
- **Dynamic Pricing:** Airlines and ride-sharing apps (like Uber) use regression models to adjust prices in real-time based on demand and supply features.

